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You Belong in the Loop

Building agentic AI workflows inside real constraints

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You have been to this session before.

A partial inventory of what has already been done to you

9:00 AM	Demystifying AI	Ballroom B
10:15 AM	AI: Threat or Opportunity?	Ballroom B
11:30 AM	Ten Prompts That Will Change How You Work	Ballroom B
1:00 PM	ChatGPT for [Your Job Title Here]	Ballroom B
2:15 PM	The Future of Work, Again	Ballroom B
3:30 PM	You Belong in the Loop: Building Agentic AI Workflows...	<i>you are here</i>

I have given two of these. I would rather not specify which two.

It isn't the technology. It's the repetition.

Three reasons those sessions didn't stick — none of which were your fault, and all of which are design problems.

REASON ONE

The prompts expired

A prompt is a fact about one model on one Tuesday. My autograder pinned temperature to zero so two runs would produce the same grade. On current models that parameter returns a 400.

REASON TWO

The demo was the product

Every one of those sessions ended at the impressive output. Not one of them showed you the part where somebody has to check it.

REASON THREE

Nothing compounded

Tactics don't stack. You started from zero every time. Six sessions in, you have six afternoons and no practice.

So this one is built backwards: one rule, one method, one page you don't have to photograph.

Hold me to it. If you leave here with a prompt, I have wasted your hour — and you have somewhere better to be, because it's 3:30 and there's a reception.

Watch this.

And watch where I stop.

What just happened

- | | | |
|----|------------------------------|--|
| 01 | I said a sentence | Out loud. Not typed. That distinction is not cosmetic and I'll come back to it. |
| 02 | It made a plan | Decomposed the request into steps I did not specify and cannot fully predict. |
| 03 | It used tools | Read files, ran queries, wrote output. Actions with consequences outside the chat. |
| 04 | It produced something | Fluent, formatted, confident. This is the part that gets demoed. |
| 05 | I checked it | This is the part that doesn't. |

THE RULE

Delegate the task whose output you can verify faster than you could produce it.

That's it. That's the session. The remaining fifty minutes are the consequences.

Run it backwards

If verification costs more than doing it yourself, delegation is a loss you haven't priced yet.

THE TASK	READS AS	WHAT THE CHECK ACTUALLY COSTS
Summarizing a document you haven't read	<i>Free time</i>	You have to read the document.
Drafting prose in your own voice	<i>A first draft</i>	Line by line — because the failures are fluent.
Code against an API you don't know	<i>Rented expertise</i>	You learn the API anyway, later, under worse conditions.
Numbers from a source you can't audit	<i>An answer</i>	Nothing. That's the problem.

Three of these are still worth delegating. You just have to book the check.



Three points on a curve

Prompt, workflow, agent. Not three tools — three positions on a single trade, with honest costs and different ways of going wrong.

Prompt, workflow, agent

PROMPT

You, in the chair

IS: one instruction, one output, every time.

COSTS: your attention, per use, forever.

FAILS: immediately and in front of you.

RIGHT WHEN: the task is rare, or the judgment is the actual product.

WORKFLOW

Fixed steps, model at named points

IS: a deterministic pipeline that calls a model where a model is needed.

COSTS: built once, maintained forever.

FAILS: loudly. It breaks and it stops.

RIGHT WHEN: the task repeats and the steps hold still.

AGENT

A goal, some tools, and latitude

IS: a system that chooses its own steps toward an outcome you specified.

COSTS: the build, plus verification that never ends.

FAILS: quietly, plausibly, and at volume.

RIGHT WHEN: the steps genuinely can't be known in advance — and you can afford the check.

Most people reach for the third when they needed the second. It's more fun to build and much harder to trust.

What moves as you hand over more

Schematic. The axis is real. The numbers are not, and I'm not going to invent them for a slide.



MIGRATION ONE

Your cost moves from doing to checking

The work does not disappear. It changes shape, loses its deadline, and stops looking like work to anyone approving your time.

MIGRATION TWO

Failure moves from loud to quiet

A broken script pages you. A confidently wrong agent gets filed, forwarded, and cited.

**An agent is a machine for making
errors plausible.**

That isn't a reason not to build one. It's the spec.



Three systems, three constraints

A regulated insurer where you cannot paste client data anywhere. A university course where the stakes are somebody's grade. A dissertation instrument that reasons over its own data.

Regulated enterprise

National Indemnity Company — cleansing, transformation, and reporting of financial data across a variety of insurance types

WHERE MOST AI ADVICE DIES

The constraint

You cannot paste client data into a public tool. The network inspects your traffic. Every automated output has to survive an audit years after you built it.

THE PATTERN

The move

Send the shape of the problem out, never the records. Schema, column names, a synthetic row that resembles nothing real. The reasoning comes back; the data never left.

What it actually looks like: SQL Server, Access, VBA, Power Query. An unglamorous stack doing unglamorous work. The model never touches a row of it — it writes the transform, and the transform runs inside the wall.

Delegable: the transform, the documentation, the test cases. **Not delegable:** the judgment about what the number means.

The autograder

Introduction to Cybersecurity — Seton Hill University

PRODUCTION, NOT DEMO

The constraint

The stakes are a student's grade. There is no version of this where a plausible-sounding error is acceptable, and the person it happens to has the least power in the system.

THE PATTERN

The move

Students submit against a Word template I wrote, as DOCX or PDF. The grader scrapes the ranges I specified, compares them to a rubric that existed first, and a strict prompt returns feedback per rubric category.

The part worth stealing: the rubric existed first. It is a written statement of what a wrong answer looks like, produced before the system could influence it. That is not a grading artifact — it is the evaluation harness, and I got it for free because teaching already required one.

Nothing goes out unread. That is not an AI rule — it is the rule for anything with my name on it.

The research instrument

Dissertation — mixed methods autoethnography, University of Nebraska—Lincoln

THE AWKWARD ONE

The constraint

The tool is both the accommodation I rely on and the object of study. I cannot pretend to stand outside it, so the method has to survive being written by an interested party.

THE PATTERN

The move

A graph-database-backed research tracker with a reasoning layer over it, on serverless infrastructure. It holds the data and it reasons about the data, and every inference is logged as a claim I can go argue with later.

Situated in: the 10(D) 8(Re) framework for agentic AI in learning design — Olmanson, Mwayaona, Seim & Stevens, EdMedia + Innovate Learning 2026, Edinburgh. The framework is on the reference card; the full citation is on its back page.

Delegable: retrieval, coding passes, surfacing the contradiction. **Not delegable:** what it means, and whether I believe it.

Three environments, one shape

ONE

The check existed first

An audit trail. A rubric. A logged claim. In all three, the thing that makes the output trustworthy was designed before the output was.

TWO

The model never held the keys

It writes the transform, drafts the feedback, surfaces the contradiction. A person signs the number, the grade, and the finding.

THREE

The constraint made it better

Not a consolation prize. The regulated version is more auditable than the unregulated one would have been, because it had to be.

None of that is a claim about which model is good. I switch models roughly as often as they ship, and the shape survives it.



The same question, read from a second position

This is the ethics section. It is seven minutes and it is not a detour — it is the engineering question asked by somebody who isn't in the room.

Two things I know

FIRST

These tools took down a barrier no institution was going to take down for me.

I dictate most of my work. That was true before any of this existed, and for years it meant my output was capped by how long I could sustain typing. It isn't anymore.

SECOND

The same systems are measurably worse at serving people they weren't built for.

Names, dialects, disabilities, edge cases. Not as a rumor — as a documented, reproducible property of the systems I am recommending you build on.

Both are true. Held side by side like that, it's balance — and balance is the most forgettable thing a talk can do.

They aren't two facts. They're one fact, read from two positions.

Accommodation by averaging

A model is a machine for being right about the middle of its training distribution.

SAME PROPERTY

Why it worked for me

Dictation and drafting are enormously well represented. My accent is the one it was measured on. The barrier came down because my work sits near the middle of the distribution.

SAME PROPERTY

Why it fails someone else

A name it has seen rarely. A dialect it has seen rarely. A body, a claim, a case it has seen rarely. The barrier stays up because their work sits in the tail.

The property that granted me access is the property that denies it to them. Not a tension. A mechanism.

This happened while I was writing this talk

WHAT I SAID

“Seton Hill”

WHAT IT TRANSCRIBED

“student health”

THE ACCOMMODATION

Why I use this one

It is the best I have found at my voice, and I chose it on the merits. That is not an endorsement. It is a dependency, and I would choose it again tomorrow.

THE FAILURE

Why that is the problem

It did not return noise. It returned English — fluent, grammatical, and contextually apt, because I was talking about teaching and students at the time.

The better it gets at my voice, the harder its mistakes are to catch. A bad recognizer hands you nonsense you would notice. A good one hands you a sentence.

Same machine. Same property. This time both positions are mine.

So it's one question, kept in two ledgers

THE ENGINEERING LEDGER

Where does it fail?

How would I know?

What does the check cost?

Who signs off?

THE ONE THAT GETS CUT FOR TIME

Who is it worse for?

How would I know?

What does that check cost?

Who pays when nobody checks?

Four questions, twice. The left column has a budget line. The right column has a volunteer.

"Human in the loop" is doing a lot of work in that sentence

Being in the loop is not the same as having the authority to stop it.

REQUIREMENT ONE

Information

You can see what the system did, not only what it produced. A summary of an action is not a record of it.

REQUIREMENT TWO

Standing

Stopping it is a normal thing to do, not an escalation with your name on it. If halting costs you politically, you are decoration.

REQUIREMENT THREE

Time

The check is on the schedule, with a name against it. If review is unpaid overtime, there is no review — there's a checkbox.

The conference theme is "You Belong." In a loop, belonging is a permission setting — and somebody configured it.

One input, two models, no oracle

The cheapest evaluation habit there is, and the only live demo that can't fail.

THE HABIT

What I actually do

Same prompt, same input, into Claude and into Gemini. I am not picking a winner. I am looking for where they disagree, because disagreement is a free map of where the task is underspecified.

THE PRICE

What it costs

One extra call and about ninety seconds. That is the entire price of finding out your workflow has been quietly relying on one model's guess.

When they agree, you've learned very little. When they diverge, there is no third system to adjudicate — and that absence is the finding, not a problem with the method.

"The model knows" was never a sentence. There is no it, and there is no knows.



Building it anyway

Patterns for the environments where the answer is "IT will never approve that" — which is usually true, and almost never the end of the conversation.

Four patterns for a locked-down shop

MOVE THE REASONING, NOT THE DATA

Send the schema, not the rows

Structure, column names, a synthetic exemplar that resembles no real person. What comes back is a transform you run inside the wall.

AUTOMATE THE BORING EIGHTY

Deterministic first, model second

Most of what looks like an AI problem is a parsing problem wearing a hat. Reserve the model for the part that genuinely needs judgment.

GET THE NO IN WRITING

Design to the policy, not around it

Ask which specific control you're tripping. "No AI" is rarely the policy; it's the summary of one. The written version has edges you can build to.

LOG IT LIKE YOU'LL BE ASKED

The audit trail is an output, not a byproduct

Input, version, model, timestamp, who approved. If you can't reconstruct a decision in two years, you don't have a system, you have a habit.

What the retrofit actually cost

THE AUTOGRADER

Waiting on a tool I never got

I was told I would have access to an established autograder. We never converged on what that would look like, and I spent the wait designing around a tool I did not have. When I stopped waiting and wrote the rubric down instead, what I needed turned out to be a template, a text scrape, and a strict prompt.

THE DISSERTATION TRACKER

A semester and a half. Then a weekend.

I fought the tracker to make it work by voice for about a semester and a half. Then I scrapped it, wrote down what it actually had to do, and rebuilt it over a weekend.

Both times I started from a known quantity and tried to bend it. Both times the known quantity was the expensive part.

The weekend is not the good news. The weekend is how I know what the semester and a half was.

And the rebuild took a weekend because I could finally say what it had to do — the rubric again, aimed at my own tools.

So build it in at the start

Three things to design for rather than retrofit. None of them costs anything much if it comes first.

INPUT

Dictation-first

If it only works typed, it doesn't work. Voice is not a demo feature — it's the difference between a tool I use and a tool I describe.

OUTPUT

Structure, not pixels

Output something a screen reader can traverse. Markdown before PDF. Headings that mean something. A table that is a table.

EVALUATION

Test on the tail

Sample your failures by group, not at random. Random sampling finds the average failure. The one you're looking for is not average.

The third one is where this section and the ethics section turn out to be the same section.



Your turn

Eleven minutes on your own work. The worksheet is in your handout — you'll leave with something you can act on *Monday*, or with a good reason not to.

The workflow audit

Worksheet, page 1 of your handout. Pick a real task you did last week — not a hypothetical.

3 MINUTES

Write down one task

The steps you actually take. Where the judgment lives. What a wrong result would look like — write that down before anything else.

Alone, in silence.

5 MINUTES

Price the check

Turn to a neighbor. For each step: could you verify a machine's version faster than doing it? Say the number out loud, in minutes.

Pairs.

3 MINUTES

One thing you found

Not your whole task. One step that turned out to be delegable, or one that looked delegable and wasn't.

Four people, out loud.

Delegate the task whose output you can verify faster than you could produce it.

Six habits that catch both kinds of failure



Write the check before the build.

If you can't say what wrong looks like, you are not ready to automate.



Keep a wrong-answer file.

Every failure, verbatim, with the input that caused it. This becomes your test suite.



Run one input through two models.

Divergence is free signal about where the task is underspecified.



Sample by group, not at random.

Random sampling finds the average failure. That is not the one that ends up in a complaint.



Version the prompt like code.

It is code. It has a blast radius and you changed it on a Friday.



Re-run the old failures every time.

That's a regression test. You already know how to do this.

Selling it upward: "We can't verify this output" and "this output is worse for a third of our customers" are the same memo. One of them gets funded — write that one.

What survives the next model release

01

The rule

Verify faster than you could produce, or book the check.

02

The audit

Steps, judgment, what wrong looks like. Fifteen minutes, any task.

03

The wrong-answer file

The only artifact here that gets more valuable over time.

04

The two ledgers

Where does it fail, and who eats it. Asked together or not at all.

None of that mentions a product name. That is the difference between this hour and the other five on the program.



You belong in the loop.

Ask for the permission setting that proves it.

Everything is already online.

The reference card, the worksheet, and these slides. Nobody has to photograph anything, and please don't feel obliged to read any of it.

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